The identity type $\mathsf{Id}\,A\,M\,N$ with Ref and J

typing, reduction, injectivity, and subject reduction

Agda development `ID/` (domain-semantics)

This note records the rules for the element-level Martin-Löf identity type added to the type-in-type theory ($\mathsf{U} : \mathsf{U}$) of the `ID/` development, and states the two metatheorems proved for it. Everything below is machine-checked in Agda (`--without-K --exact-split`) with no postulates, no holes, and no termination/positivity pragmas.

The eliminator is Martin-Löf's *original* J with a fully dependent motive. Writing the motive and base types out,

$$C : (X\,Y : A) \to \mathsf{Id}\,A\,X\,Y \to \mathsf{U}, \qquad D : (X : A) \to C\,X\,X\,(\mathsf{Ref}\,X),$$

so that $\mathsf{J}\,C\,D\,P$ at $P : \mathsf{Id}\,A\,M\,N$ has type $C\,M\,N\,P$ (the application spine $((C\,M)\,N)\,P$).

Notation. Syntactic terms and types are written with the capital and standard names of the rules below — a base type A, endpoint terms M, N, a motive C, a base term D, a proof term P, bound variables X, Y. Finite elements of the semantic domain are written with the *reserved lowercase letters* t, u, v, w, c (together with $\perp$, the universe code $\mathbb{U}$, and the codes $\mathbf{Id}(t, u, v)$ and $\mathbf{r}(w)$). We write $[\![L]\!]_\rho$ for the value of a term L in an environment ρ, and $u \lhd [\![L]\!]_\rho$ for "u is a finite approximant of that value".

Typing rules

$$\textsc{Id}\ \frac{\Gamma \vdash A : \mathsf{U} \qquad \Gamma \vdash M : A \qquad \Gamma \vdash N : A}{\Gamma \vdash \mathsf{Id}\,A\,M\,N : \mathsf{U}} \qquad\qquad \textsc{Ref}\ \frac{\Gamma \vdash A : \mathsf{U} \qquad \Gamma \vdash M : A}{\Gamma \vdash \mathsf{Ref}\,M : \mathsf{Id}\,A\,M\,M}$$

$$\textsc{J}\ \frac{\begin{array}{c}\Gamma \vdash A : \mathsf{U} \qquad \Gamma \vdash M : A \qquad \Gamma \vdash N : A \\ \Gamma \vdash C : (X\,Y : A) \to \mathsf{Id}\,A\,X\,Y \to \mathsf{U} \qquad \Gamma \vdash D : (X : A) \to C\,X\,X\,(\mathsf{Ref}\,X) \qquad \Gamma \vdash P : \mathsf{Id}\,A\,M\,N\end{array}}{\Gamma \vdash \mathsf{J}\,C\,D\,P : C\,M\,N\,P}$$

Conversion rules

$$\textsc{J-}\beta\ \frac{\Gamma \vdash A : \mathsf{U} \qquad \Gamma \vdash M : A \qquad \Gamma \vdash C : \ldots \qquad \Gamma \vdash D : \ldots}{\Gamma \vdash \mathsf{J}\,C\,D\,(\mathsf{Ref}\,M) = D\,M : C\,M\,M\,(\mathsf{Ref}\,M)}$$

J fires only on a *literal* $\mathsf{Ref}\,M$, whose type is the diagonal $\mathsf{Id}\,A\,M\,M$; both sides of J-β therefore have the *same* type $C\,M\,M\,(\mathsf{Ref}\,M)$, so the rule carries no endpoint or motive equalities and needs no injectivity. The congruences are

$$\frac{\Gamma \vdash A = A' : \mathsf{U} \qquad \Gamma \vdash M = M' : A \qquad \Gamma \vdash N = N' : A}{\Gamma \vdash \mathsf{Id}\, A\, M\, N = \mathsf{Id}\, A'\, M'\, N' : \mathsf{U}}\ \textsc{Id-cong}$$

$$\frac{\Gamma \vdash M = M' : A}{\Gamma \vdash \mathsf{Ref}\, M = \mathsf{Ref}\, M' : \mathsf{Id}\, A\, M\, M}\ \textsc{Ref-cong}$$

$$\frac{\Gamma \vdash C = C' : \ldots \qquad \Gamma \vdash D = D' : \ldots \qquad \Gamma \vdash P = P' : \mathsf{Id}\, A\, M\, N}{\Gamma \vdash \mathsf{J}\, C\, D\, P = \mathsf{J}\, C'\, D'\, P' : C\, M\, N\, P}\ \textsc{J-cong}$$

(Grey premises re-typing the components are elided; see `ID/Syntax/Typing.agda`.)

Head reduction

Reduction is untyped weak head reduction. The single-step relation $L \rightsquigarrow_1 L'$ has the following Id-clauses (alongside the usual β and application-congruence for functions):

$$\frac{}{\mathsf{J}\, C\, D\, (\mathsf{Ref}\, M) \ \rightsquigarrow_1\ D\, M}\ \textsc{J-red} \qquad \frac{P \rightsquigarrow_1 P'}{\mathsf{J}\, C\, D\, P \ \rightsquigarrow_1\ \mathsf{J}\, C\, D\, P'}\ \textsc{J-scrut}$$

J-red is the ι-step; J-scrut lets reduction proceed into the proof position so a non-canonical proof can reduce toward a Ref. Each step is a typed conversion (J-red $\mapsto$ J-β, J-scrut $\mapsto$ J-cong), and $\rightsquigarrow_1$ is deterministic.

Domain semantics of Id and J

The model is the poset of *finite elements* $t, u, v, w, c, \ldots$ of a Scott domain, with an information order $u \leq v$, a bottom $\perp$, a universe code $\mathbb{U}$, function elements, and Π-codes. The identity type adds two shapes:

$$\mathbf{Id}(t, u, v) \quad \text{(the \textit{code} of a type } \mathsf{Id}\, A\, M\, N\text{: } t \text{ codes } A,\ u \text{ codes } M,\ v \text{ codes } N\text{),}$$

$$\mathbf{r}(w) \quad \text{(a \textit{proof value}: a witness } w\text{, the model's analogue of } \lambda \text{ for functions).}$$

Semantic membership is written $u \in a$ ("the finite element u is a member of the type-code a"). The characteristic clause for the identity type is

$$\mathbf{r}(w) \in \mathbf{Id}(t, u, v) \iff w \in t \ \wedge\ w \leq u \ \wedge\ w \leq v,$$

i.e. a proof of $\mathbf{Id}(t, u, v)$ is exactly a witness of the underlying type that lies *below both endpoints* — the domain-theoretic reading of "u and v agree down to w". All mismatched shapes (e.g. $\mathbf{r}(w) \in \Pi(\ldots)$ or $\mathbf{Id}(\ldots) \in \Pi(\ldots)$) are empty, so identity codes and proof values interact only with one another.

The approximation relation $u \lhd [\![L]\!]_\rho$ ("the finite element u approximates $[\![L]\!]_\rho$") has these clauses for the new formers (for $u \neq \perp$; every term is approximated by $\perp$):

$$\begin{aligned} \mathbf{Id}(t, u, v) \lhd [\![\mathsf{Id}\, A\, M\, N]\!]_\rho &\iff t \lhd [\![A]\!]_\rho \ \wedge\ u \lhd [\![M]\!]_\rho \ \wedge\ v \lhd [\![N]\!]_\rho, \\ \mathbf{r}(w) \lhd [\![\mathsf{Ref}\, M]\!]_\rho &\iff w \lhd [\![M]\!]_\rho, \\ c \lhd [\![\mathsf{J}\, C\, D\, P]\!]_\rho &\iff \exists w.\ w \lhd [\![P]\!]_\rho \ \wedge\ \mathsf{br}(c, w), \end{aligned}$$

where the branch condition on a proof value is $\mathsf{br}(c, \mathbf{r}(w')) \iff \mathsf{fun}(w' \mapsto c) \lhd [\![D]\!]_\rho$ (and is empty when the proof value is not a Ref). So the value of $\mathsf{Id}\,A\,M\,N$ is assembled componentwise into an **Id**-code, that of $\mathsf{Ref}\,M$ into a proof value, and the value of $\mathsf{J}\,C\,D\,P$ dispatches on the proof: whenever a proof value $\mathbf{r}(w')$ approximates $[\![P]\!]_\rho$, the value $[\![D]\!]_\rho$ applied to the witness w' approximates $[\![\mathsf{J}\,C\,D\,P]\!]_\rho$ (the single-edge condition "D sends w' to c"). **The motive C does not occur in the value** — it constrains only the *type*. This is the based-J computation rule read in the model.

The logical relation and adequacy

Adequacy is proved through a Kripke logical relation between the syntax and the model. For each type it defines, at every finite element, when a term is *valid*; write $\Gamma \Vdash L : A\ [\, u \in a \,]$ for "L is valid at type A and code a, with value u". At an identity type the relation unfolds to:

$$\begin{aligned}&\Gamma \Vdash P : \mathsf{Id}\,A\,M\,N\ [\,\mathbf{r}(w) \in \mathbf{Id}(t,u,v)\,] \iff \\ &\qquad P \rightsquigarrow^* \mathsf{Ref}\,P_0 \text{ for some term } P_0 \text{ with } \Gamma \vdash P_0 = M : A \text{ and } \Gamma \vdash P_0 = N : A \text{ (both valid).}\end{aligned}$$

That is: a valid proof *head-reduces to a canonical* $\mathsf{Ref}\,P_0$ whose witness term P_0 is validly equal to both endpoints. (Because proofs are proof-irrelevant, the underlying **Id**-code carries no selection edge; the two endpoint equalities are the whole content.)

Adequacy. By induction on derivations:

$$\Gamma \vdash L : A \implies L \text{ valid}, \qquad \Gamma \vdash L = L' : A \implies L, L' \text{ validly equal.}$$

The J *case* (the based-J driver). Suppose $\Gamma \vdash \mathsf{J}\,C\,D\,P : C\,M\,N\,P$ and let c be a finite element approximating $[\![\mathsf{J}\,C\,D\,P]\!]_\rho$.

1. By the approximation clause, c is accompanied by a semantic proof value: a $\mathbf{r}(w)$ approximates $[\![P]\!]_\rho$, and $\mathbf{r}(w) \in \mathbf{Id}(t,u,v)$, so $w \le u$ and $w \le v$.
2. The induction hypothesis for the proof P at $\mathsf{Id}\,A\,M\,N$ gives, via the relation above, a head reduction $P \rightsquigarrow^* \mathsf{Ref}\,P_0$ whose witness term P_0 is validly equal to M and to N. Hence
$$\mathsf{J}\,C\,D\,P \ \rightsquigarrow^*\ \mathsf{J}\,C\,D\,(\mathsf{Ref}\,P_0) \ \rightsquigarrow_1\ D\,P_0 \qquad \text{(J-scrut}^* \text{ then J-}\beta).$$
3. *D-edge.* The induction hypothesis for the base D (a valid function), applied to the witness, makes $D\,P_0$ valid at the diagonal type $C\,P_0\,P_0\,(\mathsf{Ref}\,P_0)$.
4. *Type transport.* From $\Gamma \vdash P_0 = M : A$, $\Gamma \vdash P_0 = N : A$ and $\mathsf{Ref}\,P_0 = P$ the syntactic types are convertible, $\Gamma \vdash C\,P_0\,P_0\,(\mathsf{Ref}\,P_0) = C\,M\,N\,P : \mathsf{U}$; together with monotonicity of the type-value along the information order this carries the validity of $D\,P_0$ to the goal type $C\,M\,N\,P$.
5. *Head expansion.* Finally, validity is stable under the head reduction $\mathsf{J}\,C\,D\,P \rightsquigarrow^* D\,P_0$ of step 2 (the contractum is valid, and the redex converts to it by J-β), so $\mathsf{J}\,C\,D\,P$ is valid at $C\,M\,N\,P$. $\square$

The two-substitution (equality) version of this argument handles J-cong; it splits the two witnesses coming from the two sides and bridges them with the endpoint equalities.

Metatheorems (corollaries of adequacy)

Both results are obtained by running adequacy at the *bottom environment* and reading the logical relation off the model.

Theorem 1 (Id-injectivity, `ID/IdInjectivity.agda`)**.** *If* $\Gamma \vdash \mathsf{Id}\, A_0\, M_0\, N_0 = \mathsf{Id}\, A_1\, M_1\, N_1 : \mathsf{U}$ *then*

$$\Gamma \vdash A_0 = A_1 : \mathsf{U}, \qquad \Gamma \vdash M_0 = M_1 : A_0, \qquad \Gamma \vdash N_0 = N_1 : A_0.$$

Evaluate the given conversion at the bottom environment: it yields a validated equality at the type-code $\mathbf{Id}(\bot, \bot, \bot)$, whose stored data are exactly the component conversions between domain, left and right endpoints; a head-reduction uniqueness lemma pins the codes, giving the three conversions.

Theorem 2 (Subject reduction, `ID/SubjectReduction.agda`)**.** *If* $\Gamma \vdash L : A$ *and* $L \rightsquigarrow_1 L'$ *then* $\Gamma \vdash L' : A$.

For the J-RED step the redex is $\mathsf{J}\, C\, D\, (\mathsf{Ref}\, K)$ with $\mathsf{Ref}\, K : \mathsf{Id}\, A\, M\, N$; the contractum $D\, K$ already has type $C\, K\, K\, (\mathsf{Ref}\, K)$, retyped to the goal $C\, M\, N\, (\mathsf{Ref}\, K)$ using $K = M$ and $K = N$ extracted from $\mathsf{Ref}\, K : \mathsf{Id}\, A\, M\, N$. For J-SCRUT the result type $C\, M\, N\, P$ is transported along $P = P'$, obtained from reduction-is-conversion; the Id-typed proof needs only the diagonal J-β, never Id-injectivity. The β-step for functions inverts λ-typing through Π-injectivity, as in the base theory.

The adequacy theorems are in `ID/Adequacy/Value.agda` (`adequacySub2` / `adequacyEqSub2`); the J driver of steps 1–5 above is `ID/Adequacy/JApp.agda` (value), `JAppCross.agda` and `JAppCongr.agda` (equality).